MoodMate: Empirical Experiment Design & Evaluation Framework

> **Document Type:** Research & Experimental Evaluation Protocol
> **System Under Test:** MoodMate (Conversational AI Companion with Iso-Principle Music Recommendation)
> **Authors/Researchers:** Nikhil Yadav & Engineering Team
> **Target Scope:** Human-Computer Interaction (HCI), Affective Computing & Applied AI Evaluation

---

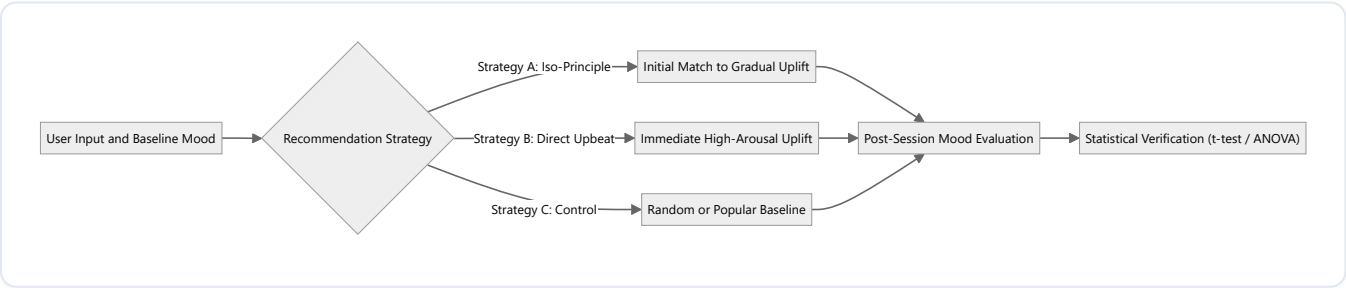
1. Executive Summary & Purpose

This document outlines the formal **experimental design** to evaluate **MoodMate**, an empathetic AI chatbot integrating conversational empathy with real-time mood detection, continuous background music playback, and mood-adaptive song curation grounded in the **iso-principle** (matching a user's current affective state before gradually transitioning toward an uplifted valence).

The goal of this evaluation is to empirically measure:
1. **Affective Efficacy:** Does the iso-principle music transition lead to statistically significant positive mood shifts compared to baseline recommendation approaches?
2. **Conversational Quality & Empathy:** How do local open-weights LLMs (Ollama Llama 3.2 3B) compare against high-parameter cloud models (Groq Llama 3.3 70B, Google Gemini 2.0 Flash) in perceived empathy, contextual understanding, and multilingual coherence?
3. **User Engagement & Agency:** How does conversational music recommendation with background streaming affect session engagement and song acceptance rates compared to static playlist recommenders?

---

2. Research Questions (RQs) & Formal Hypotheses



RQ1: Affective Regulation via the Iso-Principle

- **Question:** Does recommending songs matching the user's initial mood followed by gradual mood-elevating tracks (the iso-principle) produce superior emotional regulation compared to immediately forcing cheerful/upbeat songs?
- **Hypothesis H_1:** Participants in the **Iso-Principle condition** will demonstrate a significantly higher positive mood shift ($\Delta \text{Mood} = \text{Mood}_{\text{post}} - \text{Mood}_{\text{pre}}$) than participants presented with immediate upbeat recommendations.

- **Null Hypothesis H_0:** There is no significant difference in mood shift between iso-principle and direct upbeat recommendation strategies.

RQ2: LLM Architecture & Perceived Companion Empathy

- **Question:** What is the tradeoff between edge/local offline models (Ollama Llama 3.2 3B / Built-in Companion) and large cloud models (Llama 3.3 70B, Gemini 2.0 Flash) across perceived empathy, conversational coherence, and response latency?
- **Hypothesis H_2:** Cloud models (Groq Llama 3.3 70B / Gemini 2.0) will yield significantly higher *Bot Understanding* scores (1–5) and lower hallucination rates than compact 3B edge models, but local/built-in models will achieve significantly higher perceived *Privacy Confidence* and zero latency variance.

RQ3: Multilingual & Cultural Coherence

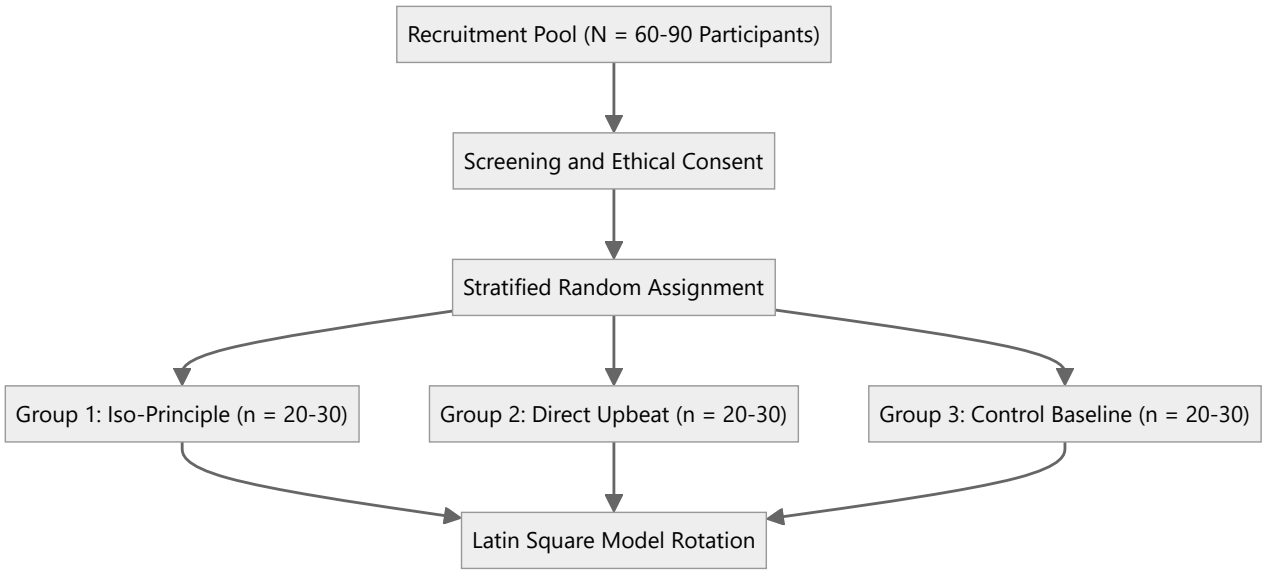
- **Question:** Does conversational mirroring in regional languages (Hindi, Bengali, and Hinglish) enhance song card click-through rates and perceived connection compared to English-only interactions?
- **Hypothesis H_3:** Users conversing in their native/preferred language (Hindi/Bengali/Hinglish) will exhibit higher song selection ratios ($S_{\text{selected}}/S_{\text{recommended}}$) and session retention when songs and explanations align culturally.

---

3. Experimental Variables

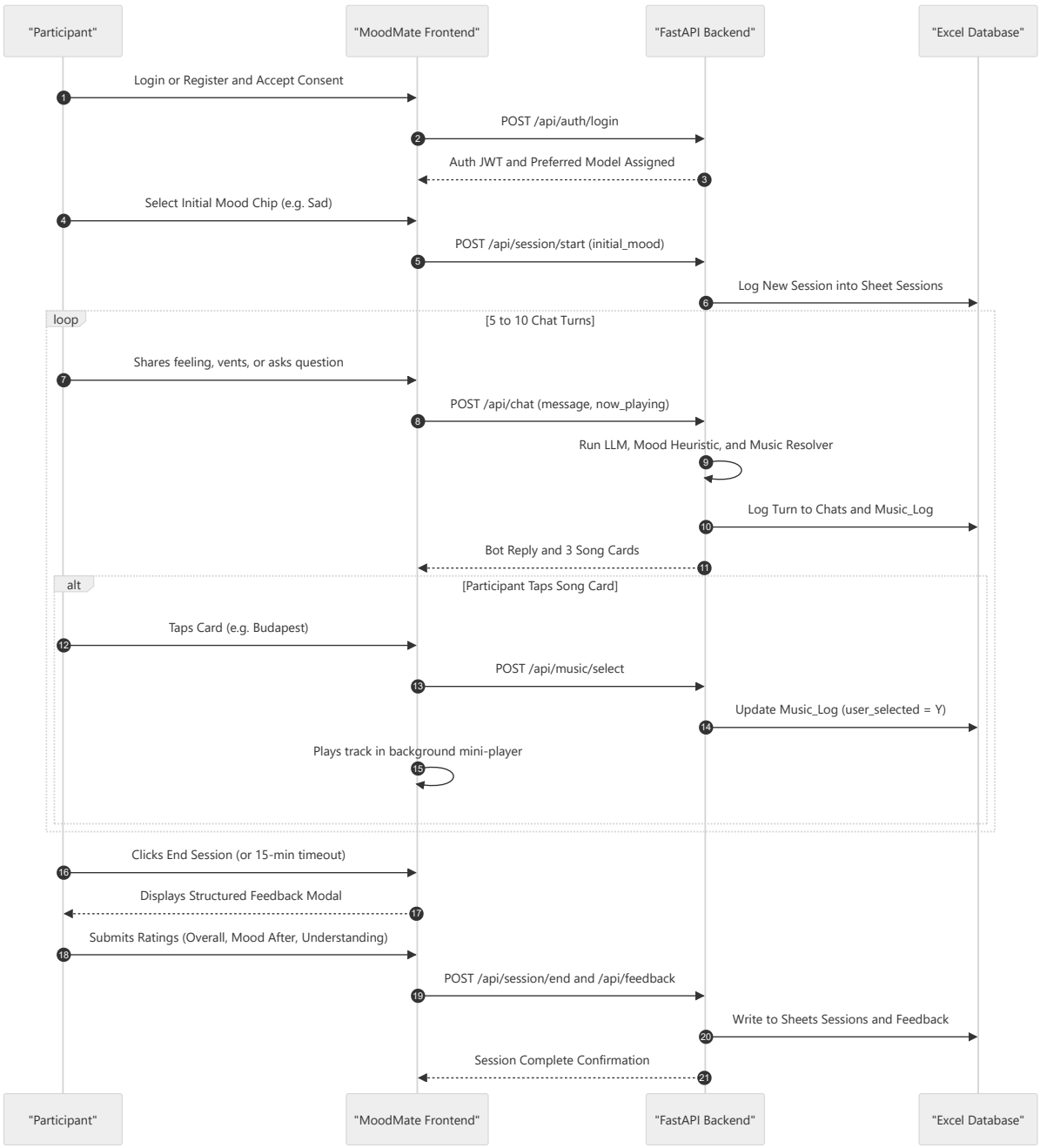
Variable Class	Variable Name	Type	Operational Definition / Measurement
Independent (IV 1)	**Music Transition Strategy**	Categorical (3 levels)	• Condition 1: **Iso-Principle** (Match → Moderate Uplift → Positive Uplift) • Condition 2: **Direct Upbeat** (Always recommend high valence/arousal songs) • Condition 3: **Control** (Popular songs without mood alignment)
Independent (IV 2)	**AI Brain Model**	Categorical (3 levels)	• Model A: `ollama/llama3.2` (Local 3B edge inference) • Model B: `groq/llama-3.3-70b-versatile` (Cloud open weights) • Model C: `gemini/gemini-2.0-flash` (Cloud multimodal LLM)
Independent (IV 3)	**Interaction Language**	Categorical (3 levels)	English, Hindi (or Hinglish), Bengali
Dependent (DV 1)	**Mood Shift (ΔMood)**	Continuous / Ordinal	Measured via Self-Assessment Manikin (SAM) / 5-point Emoji scale ($\text{Mood}_{\text{final}} - \text{Mood}_{\text{initial}}$)
Dependent (DV 2)	**Music Helpful Score**	Categorical / Ordinal	Survey response: "*Did the songs help?*" (1 = No, 2 = Somewhat, 3 = Yes)
Dependent (DV 3)	**Perceived Bot Understanding**	5-point Likert	Survey question: "*Did the bot understand you?*" (1 = Not at all, 5 = Completely)
Dependent (DV 4)	**Song Selection Rate (CTR)**	Ratio ($0.0 - 1.0$)	$\frac{\text{Count of User-Selected Songs}}{\text{Total Songs Recommended by System}}$
Dependent (DV 5)	**Session Engagement**	Quantitative	Interaction turn count, active audio playback time (minutes), overall session rating (1–5 stars)

4. Participant Cohort & Sampling Plan



1. **Cohort Size:** $N = 60$ to 90 participants (providing $> 80\%$ statistical power at $\alpha = 0.05$ for medium effect size $d = 0.5$).
2. **Inclusion Criteria:**
   - Adults (age 18–45).
   - Regular listeners of digital music (streaming at least 30 minutes daily).
   - Comfortable conversing in English, Hindi, or Bengali.
3. **Exclusion Criteria:**
   - Acute clinically diagnosed psychiatric emergencies (participants flagged by the safety system are immediately diverted to emergency support resources).
4. **Ethics & Privacy Protocol:**
   - **Informed Consent Checkbox:** Explicit participant consent recorded prior to authentication.
   - **Data Sanitization:** Participant names, phones, and emails encrypted via Fernet AES-128-CBC. Analysis performed solely on anonymized UUIDs (`user_id`).
   - **Crisis Safeguards:** Automated crisis intervention keywords immediately stop testing, display regional 24/7 helpline contacts (Vandrevala Foundation, KIRAN, AASRA), and abort recommendations.

5. Experimental Protocol & User Flow



Session Walkthrough (Step-by-Step):

1. **Pre-Session Baseline (T0):**
   - Participant logs into MoodMate.
   - Pre-session prompt asks: "*How are you feeling today?*" with standard mood chips ($1 = \text{Angry/Sad}, 5 = \text{Excited/Happy}$).
   - Initial mood is recorded in `moodmate_data.xlsx` (`Sessions` sheet).
2. **Interaction Phase (T1 – 15 Minutes):**
   - Minimum 5 conversation turns.
   - The assigned music condition triggers song recommendation cards:

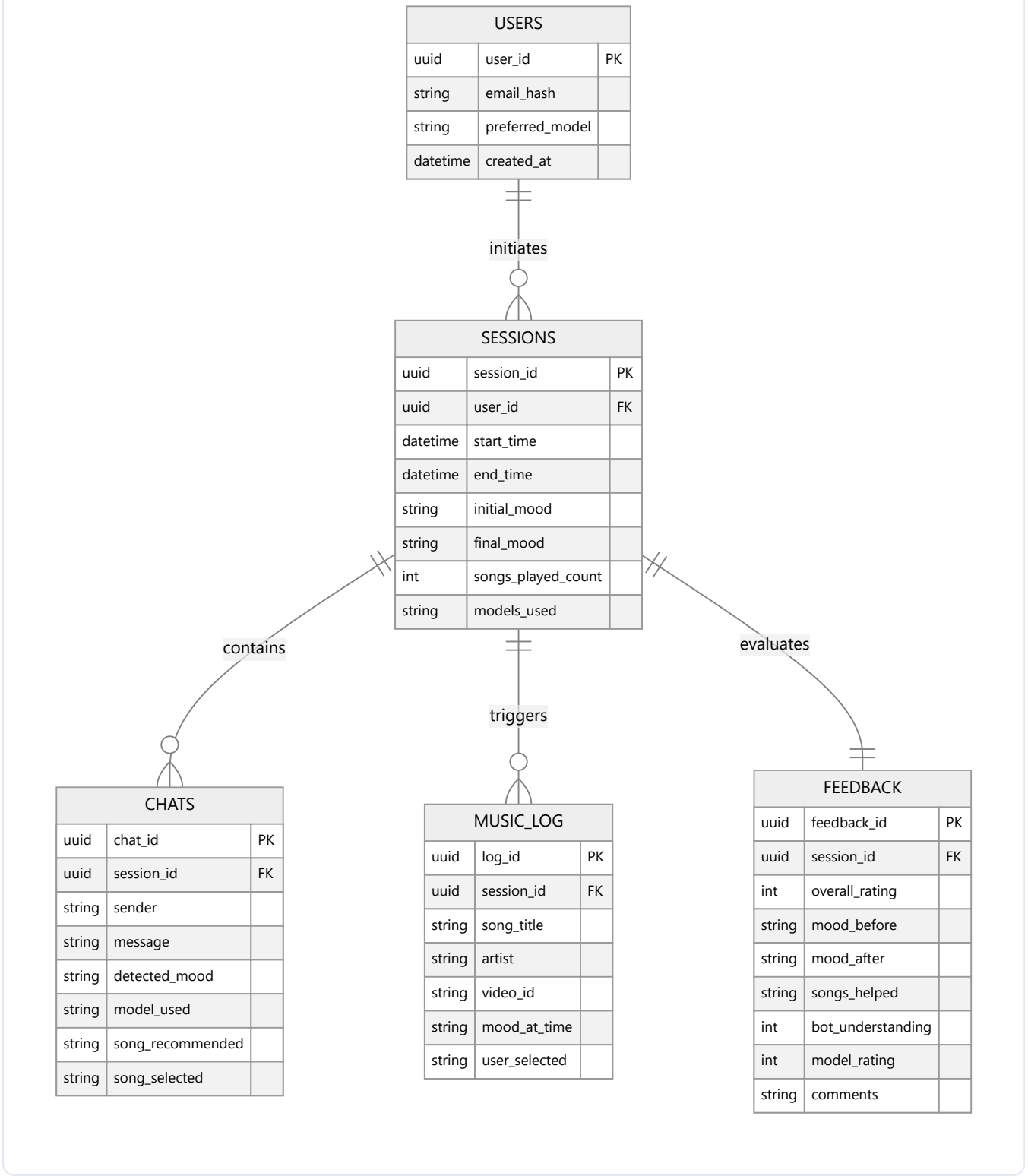
- **Group A (Iso-Principle):** First turn matches mood (e.g., low-valence calm/sad); second turn introduces mid-tempo transition; third turn introduces uplifting rhythm.
- **Group B (Direct Upbeat):** Recommends 120+ BPM upbeat party/dance tracks immediately regardless of reported sadness.
- **Group C (Control):** Recommends popular top-chart tracks randomly without mood correlation.
- Audio streams continuously via the docked mini-player while chat continues uninterrupted.

3. **Post-Session Evaluation (T2):**
   - Participant fills out the built-in feedback modal:
   - Final Mood Rating (Emoji / SAM scale).
   - Perceived Empathy / Bot Understanding (1–5 Likert).
   - Song Helpfulness (`Yes` / `Somewhat` / `No`).
   - AI Brain Rating (1–5).
   - Qualitative feedback text area.

---

6. Data Mapping to Storage Schema

All experimental telemetry is automatically persisted in real-time to `moodmate_data.xlsx` without external database dependencies:



7. Statistical Analysis & Evaluation Plan

7.1 Quantitative Analysis

1. **Mood Improvement Testing (H_1):**
   - Calculate $\Delta \text{Mood}_i = \text{Score}(\text{mood_after}_i) - \text{Score}(\text{mood_before}_i)$.

- Conduct a **One-Way Analysis of Variance (ANOVA)** (or non-parametric **Kruskal-Wallis H-test**) comparing ΔMood across Group A (Iso-Principle), Group B (Direct Upbeat), and Group C (Control).
- Post-hoc pairwise comparisons using **Tukey's HSD** or **Dunn's test** with Bonferroni correction.

2. **Model Comparison Testing (H_2):**
- Compare *Bot Understanding Score* (1–5) and *Model Rating* (1–5) between Ollama Llama 3.2, Groq Llama 3.3, and Gemini 2.0 using a **Repeated-Measures ANOVA** or **Friedman Test**.

3. **Song Engagement Metrics (H_3):**
- Compute Song Selection Click-Through Rate:
$$\text{CTR} = \frac{\sum \text{Music_Log}[\text{user_selected} == \text{'Y'}]}{\text{Count of Total Recommendations}}$$
- Pearson's χ^2 test of independence to assess whether song language preference significantly increases CTR.

7.2 Qualitative Analysis

- Thematic analysis on participant comments from the `Feedback` sheet:
- Categorizing qualitative responses into themes: *Emotional Resonance*, *Music Alignment*, *Conversational Naturalness*, and *Player Usability*.

---

8. Threats to Validity & Mitigation

Threat Category	Potential Risk	Mitigation in MoodMate Architecture
Internal Validity	Novelty effect of interacting with a musical chatbot inflating initial ratings.	Require multi-session interaction (minimum 2 sessions per participant over 48 hours).
Playback Latency	Broken YouTube links causing audio dropouts and negative bias.	**Dual-Engine Playback**: If YouTube errors (code 100/150), MoodMate automatically switches to verified 30s iTunes AAC streams or advances to the next track within 200ms.
Safety / Crisis Bias	Distress triggered by simulated crisis prompts during testing.	Server-side crisis detector checks all inputs against English, Hindi, Bengali, and Hinglish safety dictionaries *before* LLM generation, instantly offering emergency helpline guidance.
Privacy Concerns	Participants hesitating to share authentic emotional states due to fear of data leaks.	Client-side consent disclosure + master Fernet AES encryption of PII. Raw Excel inspector demonstrates that student/evaluator views cannot inspect plaintext identity without `--decrypt` flag.

---

9. Next Steps for Experiment Execution

1. **Pilot Run ($N = 6$):** Run 2 participants through each condition (Iso-Principle, Direct Upbeat, Control) to confirm logging accuracy in `moodmate_data.xlsx`.

2. **Data Export & Scrubbing:** Execute `backend/admin_view.py --raw` to verify cryptographic masking, then extract clean CSVs for R/Python analysis.
3. **Automated Analytics Script:** Generate an automated evaluation script (`evaluate_results.py`) utilizing `scipy.stats` and `pandas` to output ANOVA tables and publication-ready box plots.